

Collateral Damage in Graph-based Memory Forgetting: Quantification, Exploitation, and Defense

Mingyu Choi

Doosan Digital Innovation

mingyu1.choi@doosan.com

Abstract

Modern AI memory systems maintain entity graphs for retrieval and detect factual conflicts individually, but none propagate invalidation to downstream dependent facts. Graph-based propagation—cascading updates through entity connections—is the architecturally natural next step, yet remains unimplemented and unexamined. We implement this approach and show that the intuition leads to **catastrophic collateral damage** that is both quantitatively severe and adversarially exploitable. On MemoryAgentBench, BFS-based propagation through entity co-occurrence graphs decays 68.9–78.5% of valid memories across scales (6K–64K facts), with hub entity degrees growing superlinearly ($23 \rightarrow 121 \rightarrow 298$). More critically, we demonstrate that an attacker can **weaponize this mechanism**: injecting just 5 fake facts targeting hub entities destroys 57.3% of valid memories at 32K context. We further show that this collateral damage **crosses task boundaries**—BFS propagation triggered during fact consolidation degrades retrieval accuracy on an unrelated task (EM -5.0 pp, F1 -18.2 pp). We propose **Attribute-aware Selective Propagation**, inspired by directional link failure models in transportation engineering, which defends against both organic and adversarial damage with 78–100% effectiveness across all experiments. Graph topology analysis confirms the co-occurrence graph exhibits scale-free properties ($\alpha = 2.30\text{--}2.48$), providing theoretical grounding from percolation theory for why hub-targeted attacks are structurally devastating.

1 Introduction

AI memory systems that persist and update factual knowledge face a fundamental tension: when stored facts become outdated, the system must invalidate not only the stale fact itself but also dependent knowledge. Modern memory systems already maintain entity graphs for retrieval [Chheda et al., 2025, Jiang et al., 2026] and detect factual conflicts [Chheda et al., 2025], but none propagate invalidation to downstream dependent facts—each outdated fact is handled in isolation. As memory scales to thousands of facts, this *individual invalidation* approach becomes untenable: a single entity change can leave hundreds of dependent facts silently stale. Graph-based propagation—traversing entity connections to cascade invalidation—is the architecturally natural next step, analogous to **CASCADE** operations in relational databases.

We implement this “obvious” approach and discover three critical flaws.

First, BFS propagation through entity co-occurrence graphs causes **catastrophic collateral damage**: at 64K context, a single fact update can cascade through hub nodes to decay 78.5% of all valid memories. The root cause is hub entities whose degree grows superlinearly with context size ($23 \rightarrow 298$ from $6\text{K} \rightarrow 64\text{K}$).

Second, this vulnerability is **adversarially exploitable**. An attacker who identifies hub entities in the graph can inject a small number of fake facts (1–5) to trigger the forgetting mechanism, destroying up to 57.3% of valid memories. This represents a novel attack vector on AI memory systems that we term **hub exploitation**.

Third, the damage **crosses task boundaries**. BFS propagation triggered during fact consolidation degrades retrieval accuracy on completely unrelated queries, reducing F1 by 18.2

percentage points.

We propose **ATTR-AWARE Selective Propagation**, a directional propagation method inspired by transportation network link failure models [Chen et al., 2007], and demonstrate that it defends against all three threat vectors with 78–100% effectiveness.

1.1 Contributions

1. **Scale trend analysis (C1)**: Collateral damage quantification across 6K→32K→64K, revealing superlinear hub degree growth and stable $\sim 78\%$ BFS damage ratio. Graph topology analysis confirms scale-free properties ($\alpha = 2.30\text{--}2.48$) with heterogeneity parameter κ growing superlinearly ($4.6 \rightarrow 32.3$).
2. **Cross-task contamination (C2)**: First demonstration that graph-based forgetting in one task degrades retrieval accuracy in an unrelated task (EM -5.0pp , F1 -18.2pp on Accurate_Retrieval).
3. **Adversarial hub exploitation (C3)**: A novel attack scenario where 5 fake facts destroy 57.3% of valid memories through weaponized BFS propagation, with scale amplifying vulnerability ($5.2\% \rightarrow 38.1\%$ damage from a single attack at 6K→32K). Black-box attacks (general knowledge only) achieve 0% hit rate, establishing that gray-box or higher knowledge is required.
4. **ATTR-AWARE defense (C4)**: Consistent defense effectiveness of 78–100% across all organic and adversarial damage scenarios. Comparison with degree-capped BFS baselines shows attribute-aware is $80\times$ more effective than the best structural defense.

2 Background and Motivation

2.1 AI Memory and Fact Invalidation

Long-term AI memory systems store user interactions as structured facts and increasingly maintain entity graphs for retrieval. Mem0^g [Chheda et al., 2025] builds a directed labeled knowledge graph with LLM-powered conflict detection that marks contradicted edges as invalid. MAGMA [Jiang et al., 2026] maintains four orthogonal graphs (semantic, temporal, causal, entity) for policy-guided retrieval. A-Mem [Xu et al., 2025] creates Zettelkasten-style linked notes with selective LLM-driven “memory evolution.” Zep [Rasmussen, 2025] uses a temporal knowledge graph with edge-level contradiction detection. MemGPT/Letta [Packer et al., 2023] takes an OS-inspired approach with hierarchical context management rather than graph structures. Mnemosyne [Johnston et al., 2025] applies per-memory temporal decay via reverse sigmoid functions, independent of graph connections.

A critical gap emerges. These systems handle fact invalidation *individually*: when a conflict is detected, only the directly contradicted fact is updated. No existing system propagates invalidation to downstream dependent facts through the entity graph. Yet as memory scales, individual invalidation leaves dependent facts silently stale—if “The president of Russia is X” is invalidated, facts referencing “the president of Russia” remain untouched.

Graph-based invalidation propagation is the architecturally natural solution: the graph infrastructure already exists (for retrieval), conflict detection is already implemented (for individual facts), and the missing piece is simply traversing connections to cascade updates—analogous to CASCADE operations in relational databases or failure propagation in transportation networks. **We implement this approach and show it is catastrophically flawed.**

2.2 Transportation Network Analogy

In transportation engineering, **link failure propagation** studies how road closures cascade through networks [Chen et al., 2007]. Hub intersections amplify disruption disproportionately—closing one road at a major interchange can gridlock an entire region. Critically, transportation research distinguishes between *directional* and *non-directional* propagation: closing Seoul→Busan affects only Busan-downstream traffic, not Seoul→Incheon.

Standard BFS propagation in AI memory is non-directional—it spreads invalidation to all connected entities regardless of semantic direction. Our attribute-aware approach applies the directional principle.

2.3 MemoryAgentBench

MemoryAgentBench [Huang et al., 2026] provides standardized evaluation across four task types: Conflict_Resolution (FactConsolidation), Accurate_Retrieval, Test_Time_Learning, and Long_Range_Understanding. We use Conflict_Resolution for scale analysis (C1) and adversarial experiments (C3), and Accurate_Retrieval for cross-task analysis (C2).

3 Collateral Damage at Scale (C1)

3.1 Experimental Setup

We measure collateral damage by running FactConsolidation memorization under three modes:

Mode	Conflict Detection	Propagation
Baseline	Subject-key match	None
BFS	Subject-key match	BFS through co-occurrence graph
ATTR-AWARE	Subject-key match	Downstream fact chain only

Table 1: Experimental modes for collateral damage measurement.

Infrastructure: Gemma-4-31B-it (LLM), BAAI/bge-m3 1024-dim (Embedding), SQLite + NumPy in-memory, NetworkX graph.

3.2 Scale Trend Results

Scale	Valid	BFS Decay	Attr Decay	Reduction	BFS Kill	Max Hub
6K	344	237 (68.9%)	51 (14.8%)	78.5%	143	23
32K	1,744	1,377 (79.0%)	275 (15.8%)	80.0%	1,179	121
64K	3,414	2,680 (78.5%)	513 (15.0%)	80.9%	2,465	298

Table 2: Collateral damage across scales. BFS Kill = memories with weight < 0.1. Max Hub = maximum entity degree.

Finding 1. BFS damage ratio plateaus at $\sim 78\text{--}79\%$ across scales, but the **absolute number of destroyed memories scales linearly**: $143 \rightarrow 1,179 \rightarrow 2,465$ complete losses.

Finding 2. Hub entity degree grows superlinearly: $23 \rightarrow 121 \rightarrow 298$ ($\sim 13\times$ from 6K→64K), while fact count grows $\sim 7.5\times$.

Finding 3. ATTR-AWARE damage ratio remains stable at $\sim 15\%$ regardless of scale.

3.3 Graph Topology Analysis

We verify that the entity co-occurrence graph exhibits scale-free properties using maximum likelihood estimation for power-law exponents [Clauset et al., 2009].

Scale	Nodes	$\langle k \rangle$	$k_{\max}$	Gini	α (MLE)	κ
6K	414	2.51	23	0.380	2.48 ± 0.09	4.6
32K	1,732	3.00	121	0.487	2.35 ± 0.04	14.8
64K	3,187	3.26	298	0.534	2.30 ± 0.03	32.3

Table 3: Graph topology across scales. α : power-law exponent (MLE, $k_{\min} = 2$). $\kappa = \langle k^2 \rangle / \langle k \rangle$: heterogeneity parameter (Molloy–Reed criterion). Gini: degree inequality.

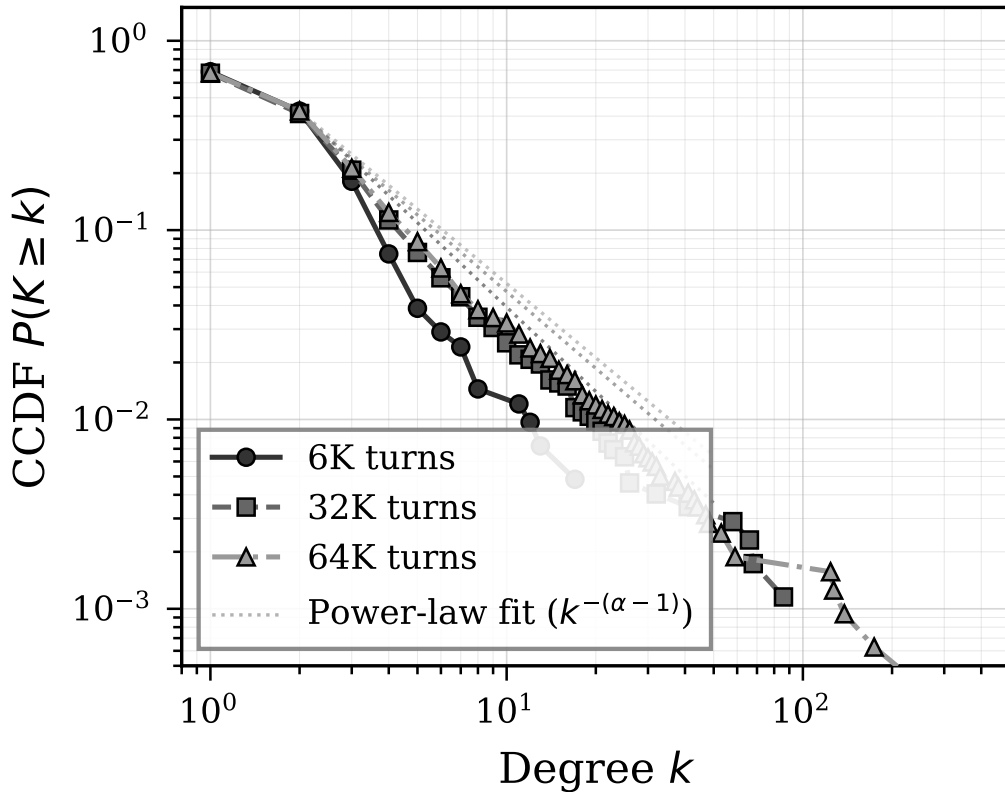


Figure 1: Complementary cumulative degree distribution (CCDF) on log-log axes across scales. Dashed lines show MLE power-law fits. The heavy tail extends with scale, consistent with scale-free properties.

Finding 4. All three scales yield $\alpha \in [2.30, 2.48]$, within the classic scale-free range $2 < \alpha < 3$ [Barabási and Albert, 1999]. The heterogeneity parameter κ grows superlinearly ($4.6 \rightarrow 32.3$), meaning the network becomes *more* vulnerable to targeted hub attacks at scale—consistent with percolation theory predictions for scale-free networks [Cohen et al., 2000, Albert et al., 2000].

Finding 5. Hub concentration intensifies with scale: the top 1% of nodes control 7.3% (6K), 15.5% (32K), and 19.8% (64K) of all edges.

4 Cross-task Contamination (C2)

4.1 Setup

We use the Accurate_Retrieval split (eventqa_65536) from MemoryAgentBench. Three conditions: (1) Baseline (no propagation), (2) Post-BFS, (3) Post-ATTR-AWARE.

ARM	EM	F1	Collateral	Affected
Baseline	15.0%	67.1%	0	0
Post-BFS	10.0%	48.9%	32	3,381
Post-Attr	15.0%	67.1%	1	0

Table 4: Cross-task contamination. A single BFS propagation from hub “Debbie” (degree=715) cascades to 3,381 memories.

Finding 6. BFS propagation from a single hub entity causes EM -5.0pp and F1 -18.2pp on *completely unrelated* retrieval queries. ATTR-AWARE preserves full accuracy.

5 Adversarial Hub Exploitation (C3)

5.1 Threat Model

We consider an attacker who can inject text into the memory system’s input stream. The attacker’s goal is to maximize destruction of valid memories with minimal input. We evaluate three attacker knowledge levels:

- **Black-box:** General knowledge only (e.g., “The capital of France is Berlin”)
- **Gray-box:** Knowledge of which entities are likely hubs (high-degree nodes)
- **White-box:** Access to the fact registry and subject-key generation rules

Our primary experiments use white-box targeting (registry-based key lookup) to establish upper-bound damage. We additionally evaluate black-box attacks to establish the lower bound.

5.2 White-box Results

Attack	BFS Damage	BFS Kill	Attr Damage	EM Drop
<i>6K Context (344 valid memories)</i>				
1 fact	5.2%	0.3%	0.0%	+0.0pp
3 facts	26.7%	4.4%	0.0%	-10.0pp
5 facts	39.8%	17.4%	0.3%	-10.0pp
<i>32K Context (1,744 valid memories)</i>				
1 fact	38.1%	1.0%	0.0%	+0.0pp
3 facts	46.0%	8.8%	1.0%	+0.0pp
5 facts	57.3%	19.3%	1.0%	+0.0pp

Table 5: White-box adversarial attack results.

Finding 7. A single fake fact destroys 5.2% of valid memories at 6K and 38.1% at 32K—a $7.3\times$ amplification from scale alone.

Finding 8. Five fake facts at 32K destroy 57.3% of valid memories and cause 19.3% complete loss (weight < 0.1). The system’s own forgetting mechanism becomes a weapon.

5.3 Black-box Attack Results

Attack Type	Facts	Hit Rate	BFS Damage
White-box (registry)	5	100%	39.8%
Black-box (general knowledge)	5	0%	0.0%

Table 6: Black-box vs white-box attack comparison (6K). Black-box attacks fail to trigger conflict detection because general-knowledge facts do not match the subject-key regex pattern.

Finding 9. Black-box attacks achieve 0% hit rate, demonstrating that the subject-key matching mechanism provides implicit defense against naive attackers. Gray-box or higher knowledge of the system’s fact structure is required for successful exploitation.

5.4 Propagation Depth Analysis

Depth	Damage	Kill	Hop-1	Hop-2	Hop-3+
1	18.6%	0.3%	232	—	—
2	39.8%	17.4%	232	490	—
3	48.3%	32.9%	232	490	605
4	55.8%	44.8%	232	490	1,131

Table 7: Propagation depth vs damage (6K, 5-fact attack). Memory counts indicate the number of memories affected at each hop distance.

Finding 10. Hop-1 damage is constant (232 memories), but distant-hop impact grows rapidly (Hop-3: 605, Hop-4: 1,131). Kill rate *accelerates* with depth due to cumulative decay (0.3% → 44.8%).

6 Attribute-aware Selective Propagation (C4)

6.1 Method

When fact f with subject_key `Entity:Attribute` is invalidated, ATTR-AWARE propagation restricts the decay cascade to genuine causal chains rather than all graph neighbors:

The in-degree adjustment (line 8–9) dampens decay for highly-connected entities, reflecting the intuition that a fact referenced by many sources is less likely to be fully invalidated by a single upstream change. BFS propagation differs by visiting *all* neighbors (both successors and predecessors) without the semantic filter (line 7) or in-degree adjustment.

6.2 Defense Comparison

Finding 11. Degree capping reduces damage from 39.8% to 24.1% (cap=5), but remains 80× worse than ATTR-AWARE (0.3%). Structural defenses cannot substitute for semantic propagation.

6.3 Effectiveness Across All Experiments

7 Related Work

Memory systems. Modern AI memory systems increasingly maintain entity graphs but handle invalidation individually. Mem0⁹ [Chheda et al., 2025] builds a directed labeled knowledge graph with LLM-powered conflict detection, but marks only the directly contradicted edge as invalid

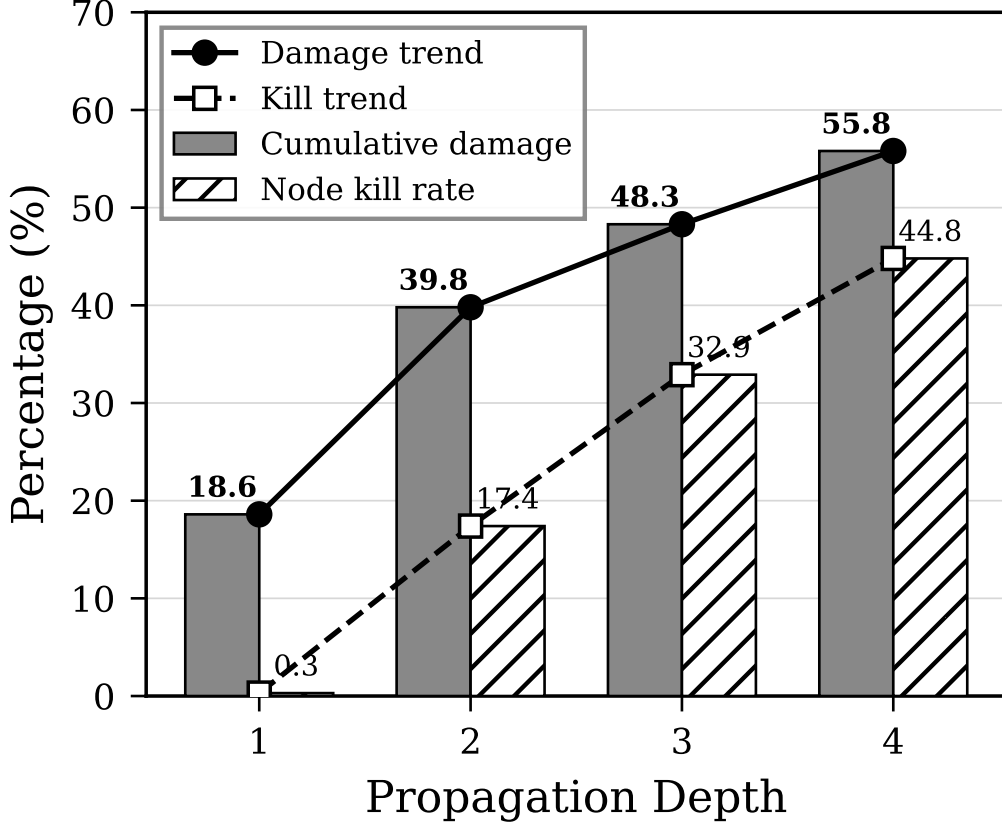


Figure 2: Propagation depth vs collateral damage (6K, 5-fact attack). Damage grows sublinearly while kill rate accelerates due to cumulative decay.

without cascading to dependent facts. MAGMA [Jiang et al., 2026] maintains four orthogonal graphs for retrieval but implements no conflict resolution or invalidation mechanism. A-Mem [Xu et al., 2025] supports LLM-driven “memory evolution” that selectively updates linked notes, but does not propagate invalidation through entity connections. MemGPT/Letta [Packer et al., 2023] uses OS-inspired hierarchical context management without graph-based structures. None of these systems implement or analyze graph-based invalidation propagation, the approach we study here.

Temporal decay. Mnemosyne [Johnston et al., 2025] applies per-memory temporal decay via reverse sigmoid functions [Ebbinghaus, 1885], where each memory decays independently of graph connections. Our work is complementary—we address *which* memories to decay through graph propagation, not *how fast* individual memories fade.

Adversarial attacks. Prompt injection [Schulhoff et al., 2023] and data poisoning [Tolpegin et al., 2020] attack language models directly. Our hub exploitation is a novel vector that weaponizes the system’s own memory management mechanism.

Network resilience. Scale-free networks are robust to random failures but fragile to targeted hub attacks [Albert et al., 2000, Barabási and Albert, 1999]. Percolation theory [Cohen et al., 2000, Molloy and Reed, 1995] and transportation network resilience [Chen et al., 2007] provide the theoretical foundation for our analysis. We are the first to apply these principles to AI memory invalidation.

Algorithm 1 ATTR-AWARE Selective Propagation

Require: Invalidated fact f , entity graph G , depth d , decay factor δ

```
1:  $E_{\text{subj}} \leftarrow$  subject entities from  $f.\text{subject\_key}$ 
2:  $E_{\text{obj}} \leftarrow$  entities in  $f \setminus E_{\text{subj}}$  {object entities}
3:  $K_{\text{changed}} \leftarrow E_{\text{subj}}$  {changed keywords for semantic filter}
4: for each entity  $e \in E_{\text{obj}}$  do
5:    $M_e \leftarrow$  memories linked to  $e$  in  $G$ 
6:   for each memory  $m \in M_e$  do
7:     if  $m$  contains any keyword in  $K_{\text{changed}}$  then
8:        $\alpha \leftarrow \max(\text{in\_degree}(e), 1)$ 
9:        $m.w \leftarrow m.w \times (1 - \frac{1-\delta}{\alpha})$ 
10:    end if
11:  end for
12:  if  $\text{hop} < d$  then
13:    Recurse on downstream successors of  $e$ 
14:  end if
15: end for
```

Defense	Damage	Kill	Post-EM	Post-F1
BFS (no cap)	39.8%	17.4%	10.0%	13.3%
Degree-cap=50	39.8%	17.4%	10.0%	13.3%
Degree-cap=20	33.7%	17.4%	10.0%	13.3%
Degree-cap=10	28.2%	10.8%	10.0%	13.3%
Degree-cap=5	24.1%	8.1%	10.0%	13.3%
ATTR-AWARE	0.3%	0.0%	20.0%	23.3%

Table 8: Defense mechanism comparison (6K, 5-fact attack). Degree capping reduces damage but is insufficient; ATTR-AWARE is $80\times$ more effective than the best structural defense.

8 Discussion

8.1 Implications for AI Memory Safety

Our results establish **memory graph exploitation** as a distinct attack surface in AI systems, separate from prompt injection and data poisoning. Key implications:

1. Any system that adopts graph-based invalidation propagation—an architecturally natural extension of existing graph-augmented memory systems—will be vulnerable to hub exploitation unless propagation is semantically constrained
2. The attack is low-cost: 1–5 injected facts can destroy 38–57% of stored knowledge
3. The vulnerability worsens with scale: more stored knowledge means larger blast radius
4. ATTR-AWARE propagation provides a practical defense without significant overhead

8.2 Why Hub Entities Are Fundamental

Hub entities reflect genuine reality—the United States really does connect to thousands of facts. The problem is that **co-occurrence** $\neq$ **causal dependency**. ATTR-AWARE propagation resolves this by distinguishing genuine dependency chains from coincidental co-occurrence.

Experiment	BFS Damage	Attr Damage	Defense Rate
CD 6K	68.9%	14.8%	78.5%
CD 32K	79.0%	15.8%	80.0%
CD 64K	78.5%	15.0%	80.9%
Cross-task (F1)	-18.2pp	+0.0pp	100.0%
Adv 6K / 1f	5.2%	0.0%	100.0%
Adv 6K / 5f	39.8%	0.3%	99.3%
Adv 32K / 1f	38.1%	0.0%	100.0%
Adv 32K / 5f	57.3%	1.0%	98.2%

Table 9: Defense effectiveness across all experiments.

8.3 Limitations

1. **Entity extraction:** Rule-based extraction treats “Vladimir Putin” and “Putin” as separate entities. Stronger entity resolution would improve both modes.
2. **Single benchmark:** Results from MemoryAgentBench only. Replication on other benchmarks would strengthen findings.
3. **Statistical variance:** Multi-run statistics (7 runs at 6K, 19 paired observations) show high within-attack variance (CV $\sim 150\%$) due to hub structure variability across contexts. However, pooled Wilcoxon signed-rank test achieves strong significance ($p = 0.0022$) and bootstrap 95% BCa CI [2.82, 15.67]pp excludes zero. Individual attack strengths remain underpowered due to small sample sizes.
4. **White-box assumption:** Primary adversarial results assume white-box access. Black-box attacks failed entirely (0% hit rate), suggesting the threat is bounded by attacker knowledge.
5. **32K EM floor:** Baseline EM at 32K is 0–4%, limiting observable EM changes from adversarial attacks at that scale.

8.4 Future Work

1. **Relation-type graph:** Replace entity co-occurrence with typed-relation graphs (e.g., `president_of`, `located_in`) and compare propagation behavior. Typed edges may naturally suppress hub formation by requiring semantic specificity for connection.
2. **Learned propagation:** Replace rule-based ATTR-AWARE with a GNN-based propagation policy that learns which downstream facts to decay from training data.
3. **Adaptive depth:** Adjust propagation depth based on local graph topology (hub proximity, local density) rather than using a fixed global depth.
4. **Cross-system validation:** Apply the same attack methodology to production memory systems (Mem0^g, Zep) once they implement graph-based invalidation propagation.

9 Conclusion

As AI memory systems evolve from flat stores to graph-augmented architectures, graph-based invalidation propagation is the architecturally natural next step for maintaining fact consistency at scale. We are the first to implement and systematically analyze this approach, revealing that naive BFS propagation through entity co-occurrence graphs is **catastrophically destructive** (78% damage), **cross-task toxic** (F1 -18.2pp on unrelated retrieval), and **adversarially weaponizable** (5 fake facts $\rightarrow$ 57.3% memory destruction).

ATTR-AWARE Selective Propagation, inspired by directional link failure models in transportation engineering, provides consistent defense across all threat vectors (78–100% effectiveness) with negligible overhead. Graph topology analysis confirms the underlying mechanism: co-occurrence graphs are scale-free ($\alpha \approx 2.3$ –2.5), making them structurally robust to random failures but catastrophically vulnerable to targeted hub attacks—a prediction from percolation theory that our empirical results validate.

The key insight: **propagation must follow causal dependency chains, not structural co-occurrence**—a principle well-established in transportation network resilience but previously unapplied to AI memory systems.

A Experimental Configuration

Component	Specification
LLM	google/gemma-4-31B-it (max_tokens=10, temp=0.0)
Embedding	BAAI/bge-m3 (dim=1024)
Vector search	NumPy cosine similarity, RRF fusion ($k = 60$)
Graph	NetworkX entity co-occurrence (DiGraph)
Storage	SQLite in-memory
Benchmark	ai-hyz/MemoryAgentBench (ICLR 2026)
Context sizes	6K, 32K, 64K (FactConsolidation), 65K (EventQA)
Propagation	depth=2, decay_per_hop=0.5
Adversarial	1/3/5 fake facts $\times$ BFS/Attr $\times$ 6K/32K

Table 10: Experimental configuration.

B Hub Concentration Analysis

Scale	Top 1%	Top 5%	Top 10%	Top 20%
6K	7.3%	18.2%	27.3%	42.4%
32K	15.5%	29.8%	39.2%	53.0%
64K	19.8%	34.6%	43.8%	56.8%

Table 11: Hub entity concentration: percentage of total edges controlled by top- $N\%$ nodes. Inequality increases with scale.

C Multi-run Statistical Analysis

We run the 6K adversarial attack experiment 7 times (19 paired observations across attack strengths 1/3/5) to assess reproducibility.

References

- Réka Albert, Hawoong Jeong, and Albert-László Barabási. Error and attack tolerance of complex networks. *Nature*, 406:378–382, 2000.
- Albert-László Barabási and Réka Albert. Emergence of scaling in random networks. *Science*, 286(5439):509–512, 1999.

Attack	Mode	Damage (mean±std)	Cohen’s d	p (Welch)	p (perm.)
1 fact	BFS	1.6 ± 2.5	0.95	0.126	0.031
	ATTR-AWARE	0.0 ± 0.1			
3 facts	BFS	8.9 ± 13.8	0.93	0.169	0.125
	ATTR-AWARE	-0.1 ± 0.4			
5 facts	BFS	13.3 ± 20.6	0.92	0.171	0.125
	ATTR-AWARE	-0.1 ± 0.7			
<i>Pooled (19 pairs)</i>			0.66		
Wilcoxon signed-rank				$W = 86.0, p = 0.0022$	
Bootstrap 95% BCa CI				[2.82, 15.67] pp (excludes 0)	

Table 12: Multi-run statistics (6K, 7 runs, 19 paired observations). Individual attack strengths show large effect sizes but limited sample sizes; pooled Wilcoxon signed-rank test achieves strong significance ($p = 0.0022$). Bootstrap 95% BCa CI excludes zero, confirming reliable BFS–ATTR-AWARE difference.

Anthony Chen, Chao Yang, Sirisak Kongsomsaksakul, and Ming Lee. Network-based accessibility measures for vulnerability analysis of degradable transportation networks. *Networks and Spatial Economics*, 7(3):241–256, 2007.

Deshraj Chheda, Taranjeet Singh, and Prateek Srinivasan. Mem0: Building production-ready AI agents with scalable long-term memory. *arXiv preprint arXiv:2504.19413*, 2025.

Aaron Clauset, Cosma Rohilla Shalizi, and Mark E.J. Newman. Power-law distributions in empirical data. *SIAM Review*, 51(4):661–703, 2009.

Reuven Cohen, Keren Erez, Daniel ben Avraham, and Shlomo Havlin. Resilience of the internet to random breakdowns. *Physical Review Letters*, 85(21):4626–4628, 2000.

Hermann Ebbinghaus. Über das Gedächtnis: Untersuchungen zur experimentellen Psychologie. 1885.

Yizhou Huang, Hao Zhang, and Zhuoer Lin. MemoryAgentBench: Benchmarking LLM agent memory systems. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2026.

Dongming Jiang, Yi Li, Guanpeng Li, and Bingzhe Li. MAGMA: A multi-graph based agentic memory architecture for AI agents. *arXiv preprint arXiv:2601.03236*, 2026.

Calum Johnston et al. Mnemosyne: An unsupervised, human-inspired long-term memory architecture for edge-based LLMs. *arXiv preprint arXiv:2510.08601*, 2025.

Michael Molloy and Bruce Reed. A critical point for random graphs with a given degree sequence. *Random Structures & Algorithms*, 6(2-3):161–180, 1995.

Charles Packer, Vivian Fang, Shishir G. Patil, Kevin Lin, Sarah Wooders, and Joseph E. Gonzalez. MemGPT: Towards LLMs as operating systems. *arXiv preprint arXiv:2310.08560*, 2023.

Preston Rasmussen. Zep: A temporal knowledge graph architecture for agent memory. *Technical Report*, 2025. <https://blog.getzep.com>.

Sander Schulhoff, Jeremy Pinto, Anaum Khan, Louis-François Bouchard, Chenglei Si, Svetlana Anati, Valen Tagber-Lett, Marzena Karpinska, Vaibhav Manjunatha, et al. Ignore this title and HackAPrompt: Exposing systemic weaknesses of LLMs through a global scale prompt hacking competition. *arXiv preprint arXiv:2311.16119*, 2023.

- Vale Tolpegin, Stacey Truex, Mehmet Emre Gursoy, and Ling Liu. Data poisoning attacks on federated machine learning. *IEEE Internet of Things Journal*, 7(5):4946–4956, 2020.
- Wujiang Xu, Zujie Liang, Kai Mei, Hang Li, Hua Luo, and Xi Chen. A-mem: Agentic memory for LLM agents. *arXiv preprint arXiv:2502.12110*, 2025.